3D MODEL · SPEC CONVERSION

E3D → PCF (CAESAR II)

Extracts E3D model to CAESAR II PCF input. Single flat MAC engine for TTY + Tkinter login GUI.

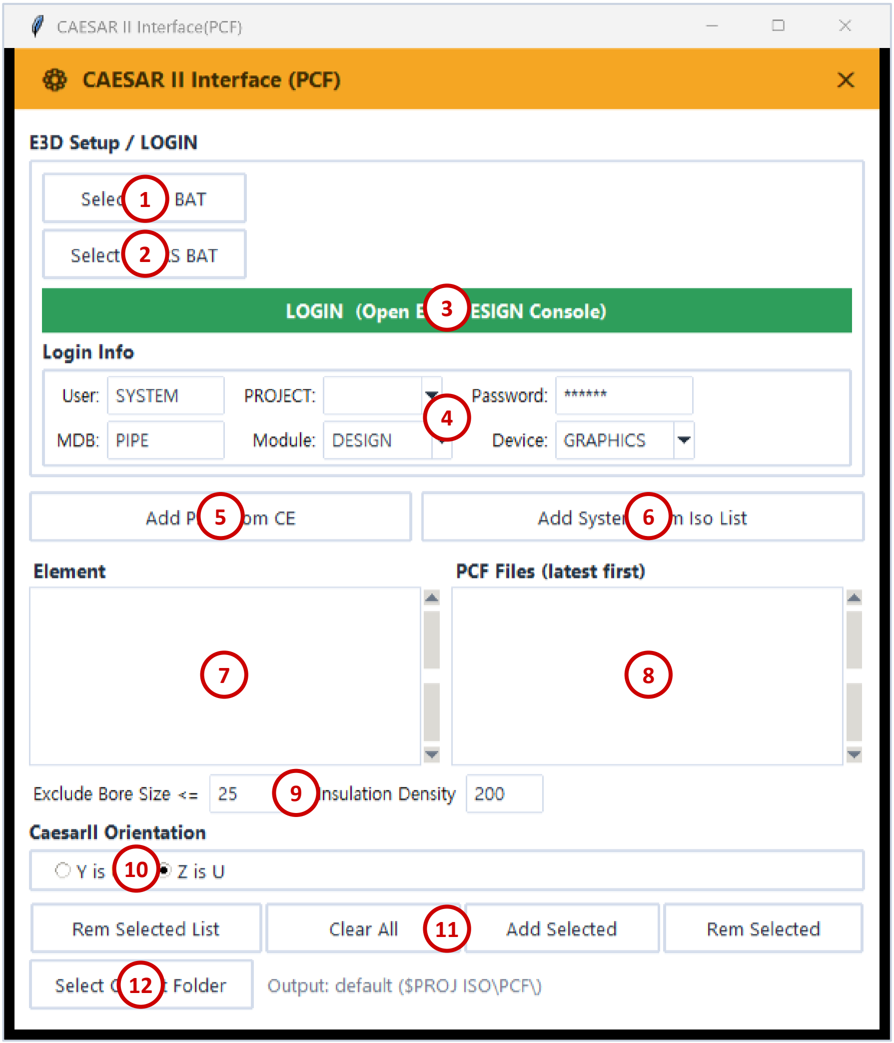
Key Purpose

Why this program

1. **Turn the live E3D model into CAESAR II input directly**
   – Selected pipes are written straight to PCF files — no manual re-modelling in the stress tool
   → **Effect:** The stress engineer gets a ready-to-import model in minutes with zero transcription error
2. **Runs inside the E3D DESIGN session as one flat MAC engine**
   – No PML form install or add-on registration — a single inlined TTY macro drives the console
   → **Effect:** Portable one-file tool that works on any project login and can later compile to a single exe
3. **Batch whole systems, not one pipe at a time**
   – Collect every PIPE under CE, or pull complete systems from the project ISO .lis list
   → **Effect:** A full line or unit is exported in one pass instead of picking components by hand
4. **Clean the model as it exports**
   – Drop small-bore noise, add insulation density, and set the CAESAR II vertical axis (Y or Z up)
   → **Effect:** The imported model is scoped and oriented correctly, with no axis flip or trivial-bore clutter

> **Required inputs :** Logged-in E3D DESIGN session (project / MDB) + PIPE(s) selected from CE or the ISO .lis system list

Main Window – Item Guide



1. Select E3D BAT (project launcher batch)
2. Select EVARS BAT (environment variables batch)
3. LOGIN — open E3D DESIGN console session
4. Login Info: User / Project / Password / MDB / Module
5. Add Pipe from CE (collect PIPEs under current element)
6. Add System from Iso List (project *.lis systems)
7. Element list — selected pipes / systems to export
8. PCF Files (generated output, latest first)
9. Exclude Bore Size <= / Insulation Density
10. CaesarII Orientation (Y is U / Z is U vertical axis)
11. Add / Rem Selected · Rem List · Clear All
12. Select Output Folder (default $PROJ ISO\PCF\)

Overview

What it does

1. **Extracts E3D model to CAESAR II PCF input**
2. **Single flat MAC engine for TTY**
3. **Tkinter login GUI**

Category: 3D Model · Spec Conversion
Folder: CAESARII
Main scripts: e3dtopcf.py
Scripts / Status: 5 scripts · Operational
Tags: E3D · PCF · CAESAR II · Stress

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: 3D Model · Spec Conversion.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: e3dtopcf.py.
- Output samples and the program window above reflect the current build.

Thank You

E3D → PCF (CAESAR II) · Plant Engineering Total Service System

Aceplant